Supplementary Material: Proofs, Algorithms and Additional Experiments for "Steering Through the Trees: Physical Lie Vector Field Evasion of Tree Ensembles in Network Intrusion Detection"

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S1. Proofs

Proposition 1 and Theorem 1 are stated in Section 3.2 of the main paper, Proposition 2 (the certificate) in Section 7.1; the notation is that of Sections 2 and 3.

Proof of Proposition 1. We evaluate the vector bracket $[X, Y] = J_Y X - J_X Y$ coordinate-wise. Items 1–5 treat the four generators of FFE-v1 (six pairs); item 6 adds the re-segmentation field and the option-padding field.

1. **Non-Derived Coordinates $\{0, 1, 2, 3, 4, 5, 6, 7, 11\}$:** For these coordinates, the components of all four generators are either constants $(0, 1, L_0, H_0)$ or depend solely on their own coordinate (e.g., $\vec{v}_{\text{win}}$ depends only on x_7). Since no generator modifies an independent variable on which another generator depends, the Jacobian products vanish identically: $(J_Y X)_k = 0$ and $(J_X Y)_k = 0$ for all $k \notin \{8, 9, 10\}$.

2. **Temporal Dilation and Payload Padding ($X = \vec{v}_{\mathbf{time}}, Y = \vec{v}_{\mathbf{pad}}$):** X has non-zero entries in rows $6, 8, 9$ ($X_6 = 1, X_8 = -x_8/x_6, X_9 = -x_9/x_6$); Y has non-zero entries in rows $4, 8, 10$ ($Y_4 = 1, Y_8 = 10^6/x_6, Y_{10} = 1/x_2$). In row 8:

$$(J_Y X)_8 = \frac{\partial Y_8}{\partial x_6} X_6 = -\frac{10^6}{x_6^2} \cdot 1 = -\frac{10^6}{x_6^2} \tag{1}$$

$$(J_X Y)_8 = \frac{\partial X_8}{\partial x_4} Y_4 + \frac{\partial X_8}{\partial x_8} Y_8 = 0 + \left(-\frac{1}{x_6}\right)\left(\frac{10^6}{x_6}\right) = -\frac{10^6}{x_6^2} \tag{2}$$

   Subtracting yields $[X, Y]_8 = -\frac{10^6}{x_6^2} - \left(-\frac{10^6}{x_6^2}\right) \equiv 0$. In rows 9 and 10, all corresponding partial derivatives vanish, so $[X, Y] = \mathbf{0}$.

3. **Temporal Dilation and Chaff Injection ($X = \vec{v}_{\mathbf{time}}, Z = \vec{v}_{\mathbf{chaff}}$):** Z has entries $Z_2 = 1, Z_4 = L_0, Z_8 = L_0 \times 10^6/x_6, Z_9 = 10^6/x_6, Z_{10} = (L_0 - x_{10})/x_2, Z_{11} = H_0$. In row 8: $(J_Z X)_8 = -L_0 \times 10^6/x_6^2 = (J_X Z)_8$. In row 9: $(J_Z X)_9 = -10^6/x_6^2 = (J_X Z)_9$. In row 10: $(J_Z X)_{10} = 0 = (J_X Z)_{10}$. Thus $[X, Z] = \mathbf{0}$.

4. **Payload Padding and Chaff Injection ($Y = \vec{v}_{\mathbf{pad}}, Z = \vec{v}_{\mathbf{chaff}}$):** In ambient $\mathbb{R}^{12}$ coordinates, $Y_{10}(x) = 1/x_2$ and $Z_{10}(x) = (L_0 - x_{10})/x_2$. Evaluating the ambient cross-Jacobian products in row 10:

$$(J_Z Y)_{10} = \sum_{k=0}^{11} \frac{\partial Z_{10}}{\partial x_k} Y_k = \frac{\partial Z_{10}}{\partial x_{10}} Y_{10} = \left(-\frac{1}{x_2}\right)\left(\frac{1}{x_2}\right) = -\frac{1}{x_2^2} \tag{3}$$

$$(J_Y Z)_{10} = \sum_{k=0}^{11} \frac{\partial Y_{10}}{\partial x_k} Z_k = \frac{\partial Y_{10}}{\partial x_2} Z_2 = \left(-\frac{1}{x_2^2}\right) \cdot 1 = -\frac{1}{x_2^2} \tag{4}$$

Subtracting gives $[Y, Z]_{10} = (J_Z Y)_{10} - (J_Y Z)_{10} = -\frac{1}{x_2^2} - \left(-\frac{1}{x_2^2}\right) \equiv 0$. In rows 8 and 9, all entries vanish identically because $Y_6 = Z_6 = 0$ while the rate entries Y_8, Z_8, Z_9 depend spatially only on duration x_6, ensuring $(J_Z Y)_k = (J_Y Z)_k = 0$ for $k \in \{8, 9\}$.

5. **Pairs Involving Window Morphing $\vec{v}_{\mathbf{win}}$:** Because $\vec{v}_{\text{win}}$ acts exclusively on x_7, which is neither read nor modified by $\vec{v}_{\text{time}}, \vec{v}_{\text{pad}}, \vec{v}_{\text{chaff}}$, all cross-Jacobians vanish identically: $[\vec{v}_k, \vec{v}_{\text{win}}] = \mathbf{0}$ for all $k \in \{1, 2, 3\}$.

6. **Re-segmentation $\vec{v}_{\mathbf{seg}} = \vec{e}_2 + \frac{10^6}{x_6}\vec{e}_9 - \frac{x_{10}}{x_2}\vec{e}_{10} + H_0\vec{e}_{11}$ (Lemma S1).** Write $S = \vec{v}_{\text{seg}}$. *With $\vec{v}_{time} = X$:* row 9 gives $(J_S X)_9 = \partial S_9/\partial x_6 \cdot X_6 = -10^6/x_6^2$ and $(J_X S)_9 = \partial X_9/\partial x_9 \cdot S_9 = (-1/x_6)(10^6/x_6) = -10^6/x_6^2$; rows 8 and 10 vanish because $S_8 = 0$, $S_6 = 0$, $X_{10} = 0$, $X_2 = 0$. *With $\vec{v}_{pad} = Y$:* row 10 gives $(J_S Y)_{10} = \partial S_{10}/\partial x_{10} \cdot Y_{10} = (-1/x_2)(1/x_2)$ and $(J_Y S)_{10} = \partial Y_{10}/\partial x_2 \cdot S_2 = -1/x_2^2$; the other rows vanish. *With $\vec{v}_{chaff} = Z$:* row 10 gives $(J_Z S)_{10} = \frac{\partial Z_{10}}{\partial x_{10}} S_{10} + \frac{\partial Z_{10}}{\partial x_2} S_2 = \frac{x_{10}}{x_2^2} - \frac{L_0 - x_{10}}{x_2^2} = \frac{2x_{10} - L_0}{x_2^2}$ and $(J_S Z)_{10} = \frac{\partial S_{10}}{\partial x_{10}} Z_{10} + \frac{\partial S_{10}}{\partial x_2} Z_2 = -\frac{L_0 - x_{10}}{x_2^2} + \frac{x_{10}}{x_2^2} = \frac{2x_{10} - L_0}{x_2^2}$; rows 8 and 9 vanish because $S_6 = S_8 = 0$ and Z_8, Z_9 depend on x_6 only. *With $\vec{v}_{win}$:* as in item 5. Finally $\vec{v}_{\text{hdr}} = \vec{e}_{11}$ is a constant field, so all its Jacobians vanish.

Hence, $[\vec{v}_i, \vec{v}_j] = \mathbf{0}$ identically for all pairs of the five generators (and for $\vec{v}_{\text{hdr}}$). □

Lemma S1 (re-segmentation). $\vec{v}_{\text{seg}}$ is the chaff field with $L_0 = 0$: it adds H_0 header bytes and one packet per unit, leaves x_4 unchanged and lowers $x_{10} = x_4/x_2$. Along $\vec{v}_{\text{seg}}$ the residual Ψ_1 is untouched ($\dot{x}_4 = \dot{x}_6 = 0$), Ψ_2 is untouched because $\dot{x}_9 - 10^6 \dot{x}_2/x_6 = 0$, and $\dot{\Psi}_3 = \dot{x}_{10} - \dot{x}_4/x_2 + x_4\dot{x}_2/x_2^2 = -\frac{x_{10}}{x_2}u_{\text{seg}} + \frac{x_4}{x_2^2}u_{\text{seg}} = -\frac{u_{\text{seg}}}{x_2}\Psi_3$. Hence the residual system of the main text holds with $A = \text{diag}(-u_{\text{time}}/x_6, -u_{\text{time}}/x_6, -(u_{\text{chaff}} + u_{\text{seg}})/x_2)$. The gap $h = x_4 - x_{11} - 20x_2$ is *not* preserved ($\dot{h} = -(H_0 + 20)u_{\text{seg}} < 0$), which is why the header-plus-payload bound g_2 of the FFE-v1 proof is not a constraint of the manifold of Section 2: the corpora's length column counts payload, and chaff is the only generator that books its wire size in it. The MTU bound $g_1 = 1500x_2 - x_4$ has $\nabla g_1 \cdot \dot{x} = 1500u_{\text{seg}} \geq 0$, and the header bounds g_3, g_4 have $\nabla g_3 \cdot \dot{x} = (H_0 - H_{\min})u_{\text{seg}} \geq 0$ and $\nabla g_4 \cdot \dot{x} = (H_{\max} - H_0)u_{\text{seg}} \geq 0$. The one new restriction is that each segmented packet carries at least one payload byte, $x_4 \geq x_2$, so $u_{\text{seg}} = 0$ on $\{x_{10} \leq 1\}$ (the integer move is $K \leq \lfloor x_4 \rfloor - x_2$).

Corollary S1 (Frobenius integrability and commuting flows)**.** *The five fields commute but are linearly dependent: $\vec{v}_{seg} = \vec{v}_{chaff} - L_0\vec{v}_{pad}$ component-wise (the entries in coordinates 4, 8 and 10 cancel). On the regular open subdomain the distribution $\Delta =$*

$\mathrm{span}\{\vec{v}_{time}, \vec{v}_{pad}, \vec{v}_{chaff}, \vec{v}_{win}\}$ *therefore has constant rank 4 on TCP (rank 3 on UDP, where* $\vec{v}_{win} = \mathbf{0}$*), and* $\vec{v}_{seg} \in \Delta$ *lies outside the nonnegative cone of chaff and padding because it needs a negative padding coefficient. Because all brackets vanish,* Δ *is involutive and integrable by Frobenius' theorem, and the flow maps commute,* $\varphi_i^{s_1} \circ \varphi_j^{s_2} = \varphi_j^{s_2} \circ \varphi_i^{s_1}$ *for* $s_1, s_2 \geq 0$*, an Abelian local Lie semigroup action on the interior of* $\widetilde{\mathcal{M}}_{flow}$*. An integer move of K segmented packets is exactly the time-K flow of* $\vec{v}_{seg}$ *(*$x_{10} = x_4/(x_2 + K)$*,* $x_{11} \to x_{11} + KH_0$*). On the interior, where the bounds are slack, the terminal state depends only on the net integrated amounts* $s_k = \int_0^T u_k\, dt$ *and not on the order of operations. At the MTU face* $g_1(x) = 1500x_2 - x_4 = 0$ *the admissibility rule requires packets to be added (chaff or segmentation) before more padding can be absorbed. The closed-form operator* $\mathcal{P}_{\mathcal{M}}$ *is a retraction onto the discrete manifold (exact on feasible inputs), not a metric projection.*

Proof of Theorem 1. Differentiating the residuals along $\dot{x} = \sum_k u_k \vec{v}_k(x)$ gives

$$\begin{aligned}\dot{\Psi}_1 &= \dot{x}_8 - \tfrac{10^6(\dot{x}_4+\dot{x}_5)}{x_6} + \tfrac{10^6(x_4+x_5)\dot{x}_6}{x_6^2} \\ &= -\tfrac{u_{\text{time}}}{x_6}\left(x_8 - \tfrac{10^6(x_4+x_5)}{x_6}\right) = -\tfrac{u_{\text{time}}}{x_6}\Psi_1, \end{aligned} \tag{5}$$

where the padding and chaff contributions $10^6(u_{\text{pad}} + L_0 u_{\text{chaff}})/x_6$ to $\dot{x}_8$ and to $10^6\dot{x}_4/x_6$ cancel, and in the same way $\dot{\Psi}_2 = -\frac{u_{\text{time}}}{x_6}\Psi_2$ and $\dot{\Psi}_3 = -\frac{u_{\text{chaff}}+u_{\text{seg}}}{x_2}\Psi_3$ (for chaff, $\dot{x}_{10} = (u_{\text{pad}} + u_{\text{chaff}}(L_0 - x_{10}))/x_2$ and $\dot{x}_4 = u_{\text{pad}} + L_0 u_{\text{chaff}}$; for segmentation, Lemma S1). Hence $\dot{\Psi} = A(t)\Psi$ with $A = \mathrm{diag}(-u_{\text{time}}/x_6, -u_{\text{time}}/x_6, -(u_{\text{chaff}}+u_{\text{seg}})/x_2)$. Since $x_6(t) \geq x_6(0) > 0$ and $x_2(t) \geq x_2(0) \geq 1$, $A \in L^1_{\text{loc}}$, and $\Psi(0) = \mathbf{0}$ because $x_0 \in \widetilde{\mathcal{M}}_{\text{flow}}$. By Carathéodory's theorem for linear systems, $\Psi \equiv \mathbf{0}$. The remaining coordinates behave as follows.

- $\dot{x}_0 = \dot{x}_1 = \dot{x}_3 = \dot{x}_5 \equiv 0$ (immutability); $\dot{x}_6 = u_{\text{time}} \geq 0$, with $x_6 \leq 1.1 \times 10^8$ by the cap rule; $\dot{x}_2 = u_{\text{chaff}} + u_{\text{seg}} \geq 0$.

- **Header bounds** $H_{\min}x_2 \leq x_{11} \leq H_{\max}x_2$: $\dot{x}_{11} = H_0(u_{\text{chaff}} + u_{\text{seg}})$ with $H_0 \in [H_{\min}, H_{\max}]$ (TCP: 20 or 32; UDP: 8), so both bounds are forward invariant; for UDP $x_{11} = 8x_2$ identically.

- **Viability.** The set is defined by the affine inequalities $g_1 = 1500x_2 - x_4 \geq 0$, $g_3 = x_{11} - H_{\min}x_2 \geq 0$, $g_4 = H_{\max}x_2 - x_{11} \geq 0$, $g_5 = x_6 - 1 \geq 0$, $g_6 = x_2 - 1 \geq 0$ and $g_7 = 1.1\times10^8 - x_6 \geq 0$ (the gap $g_2 = x_4 - x_{11} - 20x_2$ of the FFE-v1 proof is not part of the manifold, see Lemma S1; it is non-decreasing along the four FFE-v1 generators, where $\dot{g}_2 = u_{\text{pad}}$). Because every g_i is affine, the contingency cone at any boundary or corner point is the intersection of the supporting half-spaces, so it suffices that $\nabla g_i \cdot \dot{x} \geq 0$ on every active face: $\nabla g_1 \cdot \dot{x} = (1500 - L_0)u_{\text{chaff}} + 1500u_{\text{seg}} - u_{\text{pad}} \geq 0$ by the admissibility rule; $\nabla g_3 \cdot \dot{x} = (H_0 - H_{\min})(u_{\text{chaff}} + u_{\text{seg}}) \geq 0$; $\nabla g_4 \cdot \dot{x} = (H_{\max} - H_0)(u_{\text{chaff}} + u_{\text{seg}}) \geq 0$; $\nabla g_5 \cdot \dot{x} = u_{\text{time}} \geq 0$; $\nabla g_6 \cdot \dot{x} = u_{\text{chaff}} + u_{\text{seg}} \geq 0$; $\nabla g_7 \cdot \dot{x} = -u_{\text{time}} \geq 0$ on $\{x_6 = 1.1 \times 10^8\}$ because the rule forces $u_{\text{time}} = 0$ there. By the Nagumo–Brezis viability theorem (Nagumo, 1942; Brezis, 1970) the set is forward invariant.

- **Window.** For UDP $\vec{v}_{\text{win}} \equiv \mathbf{0}$ and $x_7 \equiv 0$; for TCP x_7 moves monotonically toward $W_{\text{target}} \in [0, 65535]$ without crossing it.

- **Re-segmentation.** Its only additional requirement is $x_4 \geq x_2$ (each packet carries a payload byte), which the rule $u_{\text{seg}} = 0$ on $\{x_{10} \leq 1\}$ preserves; this is a restriction on the control, not a face of $\widetilde{\mathcal{M}}_{\text{flow}}$.

- **Existence.** Denominators $x_6(t), x_2(t)$ are bounded away from zero and the fields have linear growth, so the fields are Carathéodory functions and solutions exist uniquely as absolutely continuous trajectories on $[0, \infty)$.

□

Proof of Proposition 2 (conditional certificate). The monotone-model argument is that of Incer et al. (2018). Let x' be derived from x by actions that leave x_0, x_1, x_3, x_5 unchanged, so $x'_j = x_j$ for $j \in \{0, 1, 3, 5\}$ and $x'_j \geq x_j$ for $j \in \{2, 4, 6, 11\}$. The gate G reads only the columns $\{0, 1, 2, 3, 4, 5, 6, 11\}$ and is non-decreasing in $\{2, 4, 6, 11\}$. Moving from x to x' by raising the four coordinates one at a time, each step cannot lower G (the other read coordinates are fixed along that step, including the previously raised ones), hence $G(x') \geq G(x)$, and the columns $7, 8, 9, 10$ that the actions change are not inputs of G. For an ensemble of trees with monotone constraints the property holds at the level of the summed margin, because every tree is monotone in the constrained columns along each path by construction (XGBoost, LightGBM and CatBoost enforce this) and the sum of non-decreasing functions is non-decreasing; a monotone link keeps it for probabilities. A fused alert $\max(\text{free}, G - \tau\text{-shifted})$ therefore remains raised whenever $G(x) \geq \tau$. The statement holds if the library's monotone constraint holds, which we verified by a randomized test (`tests/test_defenses_v2.py`) and did not prove for the library implementations; the verifier's absolute tolerance of 10^{-2} on the monotone columns is below the unit resolution of the integer-valued columns. The proof needs x_3 and x_5 unchanged because G reads them and is not monotone in them; it says nothing about flows with $G(x) < \tau$, and it is relative to the original flow x (a lower-footprint variant of the same attack flow is a different x). Hiding x_3 and x_5 from the gate would remove this assumption but is costly (Section S6, Tables S12 and S13): the hiding gate loses about 4 accuracy and 19 F1 points on CSE-CIC-IDS2018, flags only 74–76% of that cohort, has a benign false-positive rate of 2.1% (CSE-CIC-IDS2018) to 36% (UNSW-NB15), and V2 rebuilt on it has a whole-cohort evasion of 35–36% against FFE-v1 and 78–79% against the white-box attacker on CSE-CIC-IDS2018, against at most 9.0% for V2. □

S2. Auxiliary algorithms

Algorithm S1 is the master loop of FFE-v1, the reference baseline. In every iteration it probes the four generators once each, with the steps $\Delta s_1 = \max(1000, 0.5x_6)$ μs, $\Delta s_2 = \max(50, 0.25x_4)$ bytes, $\Delta s_3 = \max(1, \text{round}(0.5x_2))$ packets and $\Delta s_4 = |64240 - x_7|$; the directional derivatives are taken per second of delay, per byte, per packet and per window unit $\Delta s_4/65535$. It takes the step with the largest cost-normalised score decrease and falls back to boundary snapping (Algorithm S3) or a dilation of $\max(5 \times 10^4, 1.5x_6)$ μs when no probe lowers the score. With $M = 20$ iterations it uses at most $4M + 1 = 81$ queries for baseline and probes, 101 with stagnation fallbacks, plus the snapping candidates.

Algorithm S1 FFE-v1 master loop (reference baseline)

Require: Classifier F, flow $x_0 \in \mathcal{M}_{\text{flow}}$, schema $\mathcal{S}$, step budget M, costs $\vec{c}$, threshold $\tau = 0$
Ensure: Evasive flow x^* with $F(x^*) < \tau$, or FAILURE

$x \leftarrow \mathcal{P}_{\mathcal{M}}(x_0)$; $f \leftarrow F(x)$
for step $= 1$ **to** M **do**
  if $f < \tau$ **then**
    return $(x, \text{SUCCESS})$
  end if
  probe the four generators once each: $(k, \nabla_k, \vec{v}_k, \Delta s_k, f_k)$
  $k^* \leftarrow$ first maximizer of $(-\nabla_k)/c_k$ with $\nabla_k < 0$; null if none
  if $k^* \neq$ null **then**
    $x \leftarrow \mathcal{P}_{\mathcal{M}}(x + \Delta s_{k^*} \vec{v}_{k^*})$; $f \leftarrow f_{k^*}$
  else
    $x_{\text{snap}} \leftarrow \text{TSBS}(F, x, \mathcal{S})$ {Algorithm S3}
    if $F(x_{\text{snap}}) < f$ **then**
      $x \leftarrow x_{\text{snap}}$; $f \leftarrow F(x_{\text{snap}})$
    else
      $x \leftarrow \mathcal{P}_{\mathcal{M}}(x + \Delta t\, \vec{v}_{\text{time}}(x))$; $f \leftarrow F(x)$
    end if
  end if
end for
if $f < \tau$ **then**
  return $(x, \text{SUCCESS})$
else
  return $(x, \text{FAILURE})$
end if

Algorithm S4 gives the general per-round Lie-guided adversarial training scheme for tree ensembles. The experiments in Section 7 of the main paper (the LGAT baseline) run a single-round instance of it.

Algorithm S5 gives the main loop of FFE-v2 (Section 3.4 of the main paper).

S3. Diagnosis of FFE-v1 and validation tuning of FFE-v2

All diagnostics and all tuning used cohorts of up to 1,000 detected malicious flows drawn from the 10% validation split; nothing on the test cohorts was used (`results/ffev2_diagnosis.json`, `results/ffev2_tuning.json`; the numbers below that are not in these files are from `results/ffev2_report.md`, which is generated from them). On these cohorts FFE-v1 reached 99.7/99.7/58.3% (XGBoost/LightGBM/CatBoost) on CSE-CIC-IDS2018, 99.2/35.1/34.2% on CIC-IDS2017 and 60.9/64.4/72.8% on UNSW-NB15.

UNSW-NB15. On the 391 flows where FFE-v1 fails against XGBoost, the projected control evades 326. Its successful moves raise the forward packet count (99%, median +213 packets), 82% of them without changing the byte total (the projection fills the header sum at 20 B per extra packet), 7% change the duration and none the window (every TCP flow in UNSW-NB15 advertises 0 or 255 B). FFE-v1 cannot make this move: chaff packets add 28–52 B of payload each, which pulls the forward mean towards 40 B instead of down to

Algorithm S2 Analytical Manifold Projection Operator $\mathcal{P}_{\mathcal{M}}$

Require: Perturbed vector $x \in \mathbb{R}^{12}$, feature schema $\mathcal{S}$, optional baseline flow x_{orig}
Ensure: Projected flow vector $x_{\text{proj}} \in \mathcal{M}_{\text{flow}}$

$x_{\text{proj}} \leftarrow \text{copy}(x)$
if x_{orig} is provided **then**
    $x_{\text{proj}}[j] \leftarrow x_{\text{orig}}[j]$ for all $j \in \mathcal{I}_{\text{immut}}$ {Preserve immutable features}
    $x_{\text{proj}}[j] \leftarrow \max(x_{\text{orig}}[j], x_{\text{proj}}[j])$ for all $j \in \mathcal{I}^{+}_{\text{mono}}$ {Enforce monotonicity lower bounds}
end if
{Enforce discrete bounds and protocol-dependent packet/header sizes}
$x_{\text{proj}}[2] \leftarrow \max(1.0, \text{round}(x_{\text{proj}}[2]))$ {Integer Forward Packets}
$x_{\text{proj}}[3] \leftarrow x[3]$ {Immutable Backward Packets Preserved}
$H_{\min} \leftarrow$ **if** $x_{\text{proj}}[1] = 6$ **then** 20.0 **else** 8.0 {Min transport header: 20B TCP, 8B UDP}
$H_{\max} \leftarrow$ **if** $x_{\text{proj}}[1] = 6$ **then** 60.0 **else** 8.0 {Max transport header: 60B TCP, 8B UDP}
$x_{\text{proj}}[11] \leftarrow \min(x_{\text{proj}}[2] \times H_{\max}, \max(x_{\text{proj}}[2] \times H_{\min}, \text{round}(x_{\text{proj}}[11])))$ {Transport Header Length}
$x_{\text{proj}}[4] \leftarrow \min(x_{\text{proj}}[2] \times 1500.0, \max(0.0, \text{round}(x_{\text{proj}}[4])))$ {Forward Bytes $\in [0, 1500N_{\text{fwd}}]$}
$x_{\text{proj}}[5] \leftarrow x[5]$ {Immutable Backward Bytes Preserved}
$x_{\text{proj}}[6] \leftarrow \min(1.1 \times 10^8, \max(1.0, \text{round}(x_{\text{proj}}[6])))$ {Integer Duration $\geq 1\mu$s, capped below flow timeout}
$x_{\text{proj}}[7] \leftarrow$ **if** $x_{\text{proj}}[1] = 6$ **then** $\text{clamp}(\text{round}(x_{\text{proj}}[7]), 0.0, 65535.0)$ **else** 0.0 {TCP Window $[0, 65535]$, 0 for UDP}
{Analytically recalculate coupled non-linear flow rates}
$T_{\text{sec}} \leftarrow x_{\text{proj}}[6]/10^6$
$B_{\text{tot}} \leftarrow x_{\text{proj}}[4] + x_{\text{proj}}[5]$
$N_{\text{tot}} \leftarrow x_{\text{proj}}[2] + x_{\text{proj}}[3]$
$x_{\text{proj}}[8] \leftarrow B_{\text{tot}}/T_{\text{sec}}$ {Flow Bytes/s}
$x_{\text{proj}}[9] \leftarrow N_{\text{tot}}/T_{\text{sec}}$ {Flow Packets/s}
$x_{\text{proj}}[10] \leftarrow x_{\text{proj}}[4]/x_{\text{proj}}[2]$ {Forward Packet Length Mean}
return x_{proj}

3–5 B, the chaff step is capped at 50% of the packet count per iteration, padding raises the mean and dilation only lowers the rates. The score then plateaus, since several splits on the packet count, the mean length and the packet rate must be crossed jointly, and FFE-v1 spends its whole budget (89–94 queries). A random multi-start reachability test (20,000 joint draws over delay, padding, chaff, segmentation and window per failed flow, 80 failed flows per cell) finds a boundary crossing for 100% of the failed XGBoost and LightGBM flows and 97.5% of the CatBoost ones, so these failures are algorithmic, not structural.

CIC-IDS2017. FFE-v1's window target 64,240 B is almost out of support (0.08% of benign and 0.015% of attack TCP flows; 11.5% of benign flows advertise 65,535 B). For XGBoost the 8 flows FFE-v1 misses are all fixed by a window move. For LightGBM 90% of the sampled failed flows are reachable (the unreachable ones are DoS flows) and the control wins only 17 of 649 failures, all through window moves. For CatBoost only 3 of 80 sampled failed flows (3.75%) are reachable, none of the 37 DDoS and 40 DoS flows (the three are port scans): the immutable-feature ceiling of the main text; random search cannot prove unreachability. CatBoost's 22 window borders stop at 32,851.5, so 45,000, 64,240 and 65,535 share one bin and the window generator is inert for it there.

CSE-CIC-IDS2018 CatBoost. FFE-v1 evades 58.3%; the control evades 411 of the 414 failures with a pure window move. The single target 64,240 B lies on the wrong side of CatBoost's borders, whereas the in-support benign values (8,192; 65,535; 219; 237; 279;

Algorithm S3 TSBS Candidate Generation (GenerateCandidates)

Require: Current flow $x^* \in \mathcal{M}_{\text{flow}}$, active splits active_splits (pairs (j, θ_j)), tolerance ϵ_{rel}
Ensure: Candidate boundary-crossing pool $\mathcal{C}$ (before manifold projection)

$\mathcal{C} \leftarrow \emptyset$
{1. Monotonic features: closest split strictly above $x^*[j]$; UDP header x_{11} is immutable at 8B}
for each $j \in \mathcal{I}^+_{\text{mono}}$ **with** NOT $(j = 11 \wedge x^*[1] = 17)$ **do**
  $\theta^*_j \leftarrow \min\{\theta_j \mid (j, \theta_j) \in \text{active_splits} \ \wedge \ \theta_j > x^*[j]\}$
  if θ^*_j exists **then**
    $\epsilon_j \leftarrow 1.0$ **if** $j \in \{2, 4, 11\}$ **else** $\text{round}(\epsilon_{\text{rel}} \cdot \max(1.0, |\theta^*_j|))$
    $x_c \leftarrow \text{copy}(x^*)$; $x_c[j] \leftarrow \theta^*_j + \epsilon_j$
    if $j = 11$ **then**
      $x_c[11] \leftarrow \min(60.0 \cdot x_c[2], x_c[11])$
      $x_c[4] \leftarrow \min(1500.0 \cdot x_c[2], \max(0.0, x_c[4]))$
    end if
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
  end if
end for
{2. Invert coupled rate splits (x_8, x_9) into duration dilation}
for each $(r, a, b) \in \{(8, x^*[4], x^*[5]), (9, x^*[2], x^*[3])\}$ **do**
  $\theta^*_r \leftarrow \max\{\theta_r \mid (r, \theta_r) \in \text{active_splits} \ \wedge \ 0 < \theta_r < x^*[r]\}$
  if θ^*_r exists **and** $d \leftarrow 10^6(a + b)/\theta^*_r + 10.0$ satisfies $d \leq 1.1 \times 10^8$ **then**
    $x_c \leftarrow \text{copy}(x^*)$; $x_c[6] \leftarrow d$; $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
  end if
end for
{3. Invert coupled packet length splits (x_{10}) into chaff packet injection}
$H_0 \leftarrow 32.0$ **if** $(x^*[1] = 6 \wedge x^*[11]/x^*[2] \geq 26.0)$ **else** 20.0 **if** $x^*[1] = 6$ **else** 8.0
$L_0 \leftarrow 20.0 + H_0$ {52B/40B TCP (with/without RFC 7323 options), 28B UDP}
$\theta^*_{10} \leftarrow \max\{\theta_{10} \mid (10, \theta_{10}) \in \text{active_splits} \ \wedge \ L_0 < \theta_{10} < x^*[10]\}$
if θ^*_{10} exists **then**
  $N \leftarrow \lfloor (x^*[4] - \theta^*_{10} x^*[2])/(\theta^*_{10} - L_0) \rfloor + 1$
  if $N \leq 1000$ **then**
    {Chaff budget cap: $N \leq 1000$ packets}
    $x_c \leftarrow \text{copy}(x^*)$; $x_c[2] \leftarrow x^*[2] + N$
    $x_c[4] \leftarrow x^*[4] + N \cdot L_0$; $x_c[11] \leftarrow x^*[11] + N \cdot H_0$
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
  end if
end if
{4. TCP window morphing to benign OS baseline}
if $x^*[1] = 6 \ \wedge \ x^*[7] \neq 64240.0$ **then**
  $x_c \leftarrow \text{copy}(x^*)$; $x_c[7] \leftarrow 64240.0$; $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
end if
return $\mathcal{C}$

256) do not.

Where FFE-v1 stalls. (i) One hard-coded window target. (ii) One fixed magnitude per generator and iteration, too small to cross plateaus or too large for cheap flows. (iii) No generator that lowers the forward mean or raises the packet count without adding payload. (iv) Progress measured on the saturating probability, where float32 probabilities near 1 give tied probes. (v) The unit cost applied twice in the generator ranking.

Table S1 lists every validation configuration. Columns are corpus (C18 = CSE-CIC-IDS2018, C17 = CIC-IDS2017, UN = UNSW-NB15) by target (X = XGBoost, L = Light-

Algorithm S4 Lie-Guided Adversarial Training (LGAT) for Tree Ensembles

Require: Training set $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}$, number of boosting rounds T, adversarial ratio $\beta = 0.5$, step budget $M_{\text{train}} = 5$
Ensure: Robust tree ensemble $F_T(x)$
Initialize base tree $F_0 \leftarrow \text{FitCleanBaseTree}(\mathcal{D}_{\text{train}})$
for $t = 1$ **to** T **do**
  $\mathcal{D}_{\text{aug}} \leftarrow \emptyset$
  for each $(x_i, y_i) \in \mathcal{D}_{\text{train}}$ **do**
    if $y_i = 1$ **and** $\text{Uniform}(0, 1) < \beta$ **then**
      $x_{\text{adv}} \leftarrow \text{FFE}(F_{t-1}, x_i, \text{budget} = M_{\text{train}})$
      Add $(x_{\text{adv}}, 1)$ to $\mathcal{D}_{\text{aug}}$
    else
      Add (x_i, y_i) to $\mathcal{D}_{\text{aug}}$
    end if
  end for
  Compute negative gradients g_i and Hessians h_i on $\mathcal{D}_{\text{aug}}$
  Fit tree f_t using robust split scoring $\text{Gain}_{\text{robust}}$
  $F_t(x) \leftarrow F_{t-1}(x) + \eta \cdot f_t(x)$
end for
return F_T

Algorithm S5 FFE-v2 (score-only)

Require: Score oracle $p(\cdot)$ of the target, flow $x_0 \in \mathcal{M}_{\text{flow}}$ with $p(x_0) \geq 0.5$, generators $\mathcal{G} = \{\text{time}, \text{pad}, \text{chaff}, \text{seg}, \text{win}\}$, ladder ratio $r = 6$, iteration budget $M = 8$, relax-after $R = 3$, refinement probes $B = 4$, window pool $\mathcal{W}$
Ensure: Flow $x^* \in \mathcal{M}_{\text{flow}}$ with $p(x^*) < 0.5$, or FAILURE
$x \leftarrow \mathcal{P}_{\mathcal{M}}(x_0)$; $z \leftarrow \text{logit}\, p(x)$
for $t = 1$ **to** M **do**
  $\mathcal{C} \leftarrow \{\mathcal{P}_{\mathcal{M}}(\text{apply}(x, g, a)) : g \in \mathcal{G} \setminus \{\text{win}\},\ a \in \text{Ladder}_g(x)\} \cup \{\mathcal{P}_{\mathcal{M}}(x \text{ with } x_7{=}w) : w \in \mathcal{W}\}$ {duplicates removed}
  score all candidates (one query each)
  if some candidate has $p < 0.5$ **then**
    $(g, a, x') \leftarrow$ cheapest crossing candidate under $\mathcal{C}(x_0, \cdot)$; **break**
  end if
  $\Delta_c \leftarrow z - \text{logit}\, p(c)$ for each candidate; keep those with $\Delta_c \geq 10^{-3}$
  select $\arg\max\ \Delta_c / (\Delta\mathcal{C}_c + 10^{-4})$ if $t \leq R$, else $\arg\max \Delta_c$ (also if no efficient candidate exists); **stop with FAILURE if none**
  $x \leftarrow$ selected candidate; $z \leftarrow \text{logit}\, p(x)$
end for
if a crossing (g, a, x') was found **then**
  bisect a with B further probes of the parent state; keep the smallest magnitude that is still scored benign
  return x^*
end if
return FAILURE

GBM, C = CatBoost); Mean is the mean Real ASR over the nine cells, q the mean queries and Cost the geometric mean over the cells of the median operational cost. Rows from `base` to `plus_hdr` come from a first stage (a ladder of six levels from a flow-relative first step, ratio 4; each row varies one factor); `span` rows use the final span-based ladder; `refine` and

`A ratio6` rows the final design. The two `frozen` rows are the configurations evaluated once on the test cohorts, shown with FFE-v1, the control and the generic-window variant on the same validation cohorts.

Reading the table: the window pool is decisive (`win_k=0`: CSE-CIC-IDS2018 35–61%, CIC-IDS2017 XGBoost 35%); chaff matters on UNSW-NB15 for LightGBM and CatBoost; a ladder that is too short or too fine fails on UNSW-NB15 (`ratio=2`, `levels=4`: 69–82%); re-segmentation alone is worth little once chaff and padding exist; and CIC-IDS2017 LightGBM (94.9%) and CatBoost (36.3%) sit at the reachability ceiling in every configuration. Beam search (width 2 or 4: +0.1 point for 15–45 more queries), restarts (subsumed by the relax fallback), compound pair moves (the iteration already composes moves) and option padding (no gain) were not kept. Bisection refinement roughly halves the geometric-mean cost (1.07×10^{-2} to 6.3×10^{-3}) for about two queries.

S4. FFE-v2 test results in detail

Table S2 gives, for each of the nine cells, the mean queries of the control and of FFE-v2 with snapping, the median operational cost of the successful flows of each method, and exact McNemar counts (FFE-v2-only/other-only flows, p) of score-only FFE-v2 against FFE-v1, the control and CAPGD (XGBoost only). Table S3 gives paired medians on the flows that both methods evade. Table S4 gives the share of FFE-v2's successful flows that use each generator, and Table S5 the per-family Real ASR (number of flows in brackets). On CSE-CIC-IDS2018 the three libraries are evaded in 98.7%–100.0% of the flows of each family; the five missed flows are DDoS. On CIC-IDS2017 DoS is evaded in 100.0% (XGBoost), 90.5% (LightGBM) and 5.0% (CatBoost), and DDoS in 100.0%, 100.0% and 0.0%. On UNSW-NB15 the weakest large family is DoS for XGBoost (93.8%) and Fuzzers for LightGBM (94.1%).

S5. Multi-seed evaluation

Each seed retrains the targets on a new stratified split and redraws a cohort of up to 300 flows detected by XGBoost; LightGBM and CatBoost are attacked on the part of that cohort they also detect, and the frozen configuration is used without tuning. Table S6 pools the seeds: for each comparison it gives the number of flows that FFE-v2 evades and the other method does not, and vice versa, the pooled exact McNemar p, and the number of seeds in which FFE-v2's Real ASR was higher (W), equal (T); FFE-v2 was never lower in any seed.

S6. Defense details

Recipe. The gate is trained on the schema columns without the window, the two rates and the mean length, with monotone constraints +1 on forward packets, forward bytes, duration and header bytes and larger capacity (XGBoost 300 trees of depth 8; LightGBM 300 trees of depth 8 with 200 leaves; CatBoost 300 iterations of depth 8). The orbit set has 100,000 transformed rows; adversarial training uses two rounds of 3,000 crafted flows, alternating

Table S1. Validation-split tuning of FFE-v2 (Real ASR %, nine cells).

Configuration	C18-X	C18-L	C18-C	C17-X	C17-L	C17-C	UN-X	UN-L	UN-C	**Mean**	q	**Cost**
base	99.7	99.7	99.7	100.0	94.9	36.3	96.7	97.5	97.4	91.3	39.0	9.0e-03
ratio=2	99.7	99.7	99.7	100.0	94.8	36.3	68.7	70.5	69.2	82.1	57.0	5.7e-03
ratio=3	99.7	99.7	99.7	100.0	94.8	36.3	93.4	93.1	94.5	90.1	42.4	6.8e-03
ratio=6	99.7	99.7	99.7	100.0	94.9	36.3	97.9	97.8	98.1	91.6	36.4	1.1e-02
ratio=8	99.7	99.7	99.7	100.0	94.9	36.3	97.5	96.3	98.8	91.4	34.8	9.7e-03
levels=4	99.7	99.7	99.7	100.0	94.8	36.3	76.8	74.8	81.5	84.8	38.7	5.9e-03
levels=8	99.7	99.7	99.7	100.0	94.9	36.3	97.3	98.1	98.7	91.6	43.2	9.0e-03
levels=10	99.7	99.7	99.7	100.0	94.9	36.3	97.3	98.3	98.6	91.6	46.7	9.0e-03
beam=2	99.7	99.7	99.7	100.0	94.9	36.3	97.0	98.4	98.1	91.5	54.2	9.0e-03
beam=4	99.7	99.7	99.7	100.0	94.9	36.3	97.5	99.0	98.6	91.7	83.8	9.0e-03
lambda=0.0	99.7	99.7	99.7	100.0	94.8	36.3	96.7	97.7	98.9	91.5	36.9	9.0e-03
lambda=0.5	99.7	99.7	99.7	100.0	94.9	36.3	96.8	97.6	97.7	91.4	37.7	9.0e-03
lambda=2.0	99.7	99.7	99.7	100.0	94.9	36.3	96.6	97.5	97.5	91.3	39.6	9.0e-03
relax=0	99.7	99.7	99.7	100.0	94.8	36.3	96.7	97.7	98.9	91.5	36.9	9.0e-03
relax=1	99.7	99.7	99.7	100.0	94.8	36.3	96.6	97.4	97.9	91.3	37.1	9.0e-03
relax=6	99.7	99.7	99.7	100.0	94.9	36.3	96.7	97.5	97.5	91.3	41.5	9.0e-03
relax=99	99.7	99.7	99.7	100.0	94.9	36.3	96.7	97.5	97.5	91.3	41.5	9.0e-03
win_k=0	60.8	52.2	35.3	35.0	35.0	35.4	96.7	97.5	97.4	60.6	56.5	1.8e-02
win_k=3	99.7	99.7	99.7	99.6	94.9	36.1	96.7	97.5	97.4	91.2	34.0	1.0e-02
win_k=12	99.7	99.7	99.7	100.0	94.9	36.3	96.7	97.5	97.4	91.3	47.5	7.0e-03
iter=4	99.7	99.7	99.7	100.0	94.8	36.3	95.0	97.4	96.8	91.0	34.7	9.0e-03
iter=12	99.7	99.7	99.7	100.0	94.9	36.3	96.7	97.5	97.6	91.3	39.0	9.0e-03
iter=20	99.7	99.7	99.7	100.0	94.9	36.3	96.7	97.5	97.6	91.3	39.0	9.0e-03
snap_pf=4	99.7	99.7	99.7	100.0	94.9	36.3	97.2	98.6	98.5	91.6	41.6	9.0e-03
snap_pf=8	99.7	99.7	99.7	100.0	94.9	36.4	97.4	98.7	98.5	91.7	43.9	9.0e-03
snap_pf=16	99.7	99.7	99.7	100.0	94.9	36.4	98.1	98.8	98.5	91.7	54.1	9.0e-03
no_seg	99.7	99.7	99.7	100.0	94.8	36.3	96.6	96.9	97.2	91.2	32.2	9.1e-03
no_chaff	99.7	99.7	99.7	100.0	94.9	36.3	95.4	59.9	63.9	83.3	39.5	6.3e-03
no_time	99.7	99.7	99.7	100.0	94.8	36.3	95.3	93.3	98.7	90.8	30.4	1.5e-02
no_pad	99.7	99.7	99.7	100.0	94.9	36.3	94.5	97.0	97.0	91.0	30.5	9.1e-03
plus_hdr	99.7	99.7	99.7	100.0	94.9	36.3	96.9	97.6	97.5	91.4	45.0	9.0e-03
span ratio=3	99.7	99.7	99.7	100.0	94.9	36.3	97.2	98.6	98.8	91.6	55.2	9.3e-03
span ratio=4	99.7	99.7	99.7	100.0	94.9	36.3	97.4	98.0	98.4	91.6	46.4	9.2e-03
span ratio=6	99.7	99.7	99.7	100.0	94.9	36.3	98.2	97.7	98.2	91.6	39.4	1.1e-02
span ratio=8	99.7	99.7	99.7	100.0	94.9	36.3	97.4	95.0	98.6	91.2	36.3	9.3e-03
span ratio=12	99.7	99.7	99.7	100.0	94.9	36.3	97.4	94.7	98.0	91.1	33.1	1.7e-02
span ratio=16	99.7	99.7	99.7	100.0	94.9	36.3	97.2	96.1	98.0	91.3	31.3	1.8e-02
span pad_min=1	99.7	99.7	99.7	100.0	94.9	36.3	98.4	97.3	98.4	91.6	40.6	1.1e-02
span pad_min=16	99.7	99.7	99.7	100.0	94.9	36.3	98.2	97.7	98.2	91.6	39.4	1.1e-02
span pad_min=100	99.7	99.7	99.7	100.0	94.9	36.3	98.2	97.7	98.3	91.6	38.0	1.1e-02
span time_min=100	99.7	99.7	99.7	100.0	94.9	36.3	97.7	97.2	98.0	91.5	41.4	1.1e-02
span time_min=10000	99.7	99.7	99.7	100.0	94.9	36.3	98.2	97.4	97.9	91.5	39.1	8.3e-03
span pkt_max=512	99.7	99.7	99.7	100.0	94.9	36.3	96.7	97.4	97.3	91.3	38.5	1.1e-02
span pkt_max=16384	99.7	99.7	99.7	100.0	94.9	36.3	98.2	98.0	98.4	91.6	41.0	1.1e-02
A ratio6	99.7	99.7	99.7	100.0	94.9	36.3	98.2	97.7	98.2	91.6	39.4	1.1e-02
refine=3 ratio=3	99.7	99.7	99.7	100.0	94.9	36.3	97.2	98.6	98.8	91.6	56.7	5.9e-03
refine=6 ratio=3	99.7	99.7	99.7	100.0	94.9	36.3	97.2	98.6	98.8	91.6	58.1	5.6e-03
refine=3 ratio=4	99.7	99.7	99.7	100.0	94.9	36.3	97.4	98.0	98.4	91.6	47.9	6.3e-03
refine=6 ratio=4	99.7	99.7	99.7	100.0	94.9	36.3	97.4	98.0	98.4	91.6	49.2	6.0e-03
refine=3 ratio=6	99.7	99.7	99.7	100.0	94.9	36.3	98.2	97.7	98.2	91.6	40.8	6.6e-03
refine=6 ratio=6	99.7	99.7	99.7	100.0	94.9	36.3	98.2	97.7	98.2	91.6	42.1	6.1e-03
refine=3 ratio=8	99.7	99.7	99.7	100.0	94.9	36.3	97.4	95.0	98.6	91.2	37.7	6.5e-03
refine=6 ratio=8	99.7	99.7	99.7	100.0	94.9	36.3	97.4	95.0	98.6	91.2	39.1	6.4e-03
refine=4 pkt16k	99.7	99.7	99.7	100.0	94.9	36.3	98.2	98.0	98.4	91.6	42.9	6.3e-03
refine=4 beam2	99.7	99.7	99.7	100.0	94.9	36.3	98.9	98.6	97.9	91.7	56.5	6.2e-03
refine=4 snap8	99.7	99.7	99.7	100.0	94.9	36.3	98.7	98.6	98.7	91.8	46.1	6.3e-03
refine=4 snap16 beam2	99.7	99.7	99.7	100.0	94.9	36.4	99.1	99.6	98.5	91.9	80.6	6.3e-03
refine=4 plus_hdr	99.7	99.7	99.7	100.0	94.9	36.3	98.2	97.8	98.1	91.6	47.2	6.2e-03
FFE-v2 frozen (score-only)	99.7	99.7	99.7	100.0	94.9	36.3	98.2	97.7	98.2	91.6	41.3	6.3e-03
FFE-v2 frozen (+snapping)	99.7	99.7	99.7	100.0	94.9	36.3	98.7	98.6	98.7	91.8	46.1	6.3e-03
FFE-v1	99.7	99.7	58.3	99.2	35.1	34.2	60.9	64.4	72.8	69.4	59.7	8.2e-02
White-box control (project in loop)	99.7	99.7	99.7	100.0	36.8	36.3	91.4	87.3	57.8	78.7	2.0	1.8e-02
FFE-v2 generic windows	99.7	99.7	99.7	99.6	94.9	36.3	98.2	97.7	98.2	91.5	40.6	8.7e-03

Table S2. Queries, median operational cost of successful flows and exact McNemar tests (FFE-v2-only/other-only, p).

Corpus	Target	q ctrl	q snap	cost v1	cost ctrl	cost v2	v2 vs v1	v2 vs ctrl	v2 vs CAPGD
CSE-CIC-IDS2018	XGBoost	2.0	30.8	0.00057	0.000495	3.02e-05	0/0, 1	0/0, 1	110/0, 1.5×10^{-33}
	LightGBM	2.0	31.6	0.000977	0.000496	2.42e-05	0/0, 1	0/0, 1	–
	CatBoost	2.0	29.2	0.000977	0.000285	0.000285	402/0, 1.9×10^{-121}	0/0, 1	–
CIC-IDS2017	XGBoost	2.0	28.3	0.000855	0.000376	0.000554	8/0, 0.0078	0/0, 1	105/0, 4.9×10^{-32}
	LightGBM	2.0	36.5	1.62	1.56e-05	0.000554	609/0, 9.4×10^{-184}	594/0, 3.1×10^{-179}	–
	CatBoost	2.0	140.4	0.5	0.00671	3.22e-06	16/0, 3.1×10^{-5}	0/0, 1	–
UNSW-NB15	XGBoost	2.0	37.9	5.61	213	40	412/5, 6.1×10^{-115}	89/8, 2.0×10^{-18}	772/0, 8.1×10^{-233}
	LightGBM	2.0	41.5	7.49	49	50.4	349/6, 7.4×10^{-95}	107/21, 4.5×10^{-15}	–
	CatBoost	2.0	42.3	9.8	10.5	56	270/1, 1.4×10^{-79}	400/0, 7.7×10^{-121}	–

Table S3. Paired median operational cost (FFE-v2 / other method) on the flows both evade.

Corpus	Target	v2 / v1 (flows both evade)	v2 / control (flows both evade)
CSE-CIC-IDS2018	XGBoost	3.02e-05 / 0.00057 (n=995)	3.02e-05 / 0.000495 (n=995)
	LightGBM	2.42e-05 / 0.000977 (n=995)	2.42e-05 / 0.000496 (n=995)
	CatBoost	0.000121 / 0.000977 (n=593)	0.000285 / 0.000285 (n=995)
CIC-IDS2017	XGBoost	0.000554 / 0.000855 (n=992)	0.000554 / 0.000376 (n=1000)
	LightGBM	1.17e-05 / 1.62 (n=351)	1.17e-05 / 1.56e-05 (n=366)
	CatBoost	3.22e-06 / 0.5 (n=347)	3.22e-06 / 0.00671 (n=363)
UNSW-NB15	XGBoost	6.72 / 5.77 (n=571)	32.5 / 213 (n=894)
	LightGBM	14.6 / 7.49 (n=618)	35.8 / 49 (n=860)
	CatBoost	29.1 / 9.78 (n=688)	22.1 / 10.5 (n=558)

Table S4. Share (%) of FFE-v2's successful flows that use each generator.

Corpus	Target	time	pad	chaff	seg	win
CSE-CIC-IDS2018	XGBoost	31	0	0	0	69
	LightGBM	49	0	2	0	49
	CatBoost	2	0	0	0	98
CIC-IDS2017	XGBoost	12	0	0	0	87
	LightGBM	14	1	0	1	87
	CatBoost	79	0	6	0	15
UNSW-NB15	XGBoost	14	13	37	39	0
	LightGBM	12	14	64	16	0
	CatBoost	11	7	61	24	0

Table S5. Real ASR (%) of FFE-v2 by attack family (number of flows).

Corpus	**Family**	**XGBoost**	**LightGBM**	**CatBoost**
CSE-CIC-IDS2018	DoS	100.0 (390)	100.0 (390)	100.0 (390)
	DDoS	98.7 (375)	98.7 (375)	98.7 (375)
	BruteForce	100.0 (200)	100.0 (200)	100.0 (200)
	Botnet	100.0 (35)	100.0 (35)	100.0 (35)
CIC-IDS2017	DoS	100.0 (422)	90.5 (422)	5.0 (420)
	PortScan	100.0 (322)	100.0 (322)	100.0 (321)
	DDoS	100.0 (235)	100.0 (235)	0.0 (235)
	BruteForce	100.0 (20)	100.0 (20)	100.0 (20)
	Botnet	100.0 (1)	100.0 (1)	100.0 (1)
UNSW-NB15	Exploits	99.8 (504)	98.0 (504)	97.2 (497)
	Fuzzers	95.9 (219)	94.1 (219)	98.1 (213)
	Reconnaissance	100.0 (143)	96.5 (143)	96.5 (143)
	DoS	93.8 (80)	95.0 (80)	97.4 (77)
	Generic	90.9 (22)	100.0 (22)	100.0 (21)
	Shellcode	100.0 (19)	100.0 (19)	100.0 (19)
	Backdoor	100.0 (6)	100.0 (6)	100.0 (6)
	Worms	100.0 (6)	83.3 (6)	100.0 (6)
	Analysis	100.0 (1)	100.0 (1)	100.0 (1)

Table S6. Pooled paired comparison of FFE-v2 with FFE-v1, the control and CAPGD over seeds.

Corpus	**Target**	**vs FFE-v1**	**vs control**	**vs CAPGD**
CSE-CIC-IDS2018	XGBoost	0/0, 1 (0W 5T)	0/0, 1 (0W 5T)	30/0, 1.9×10^{-9} (4W 1T)
	LightGBM	0/0, 1 (0W 3T)	0/0, 1 (0W 3T)	7/0, 0.016 (2W 1T)
	CatBoost	388/0, 3.2×10^{-117} (3W 0T)	0/0, 1 (0W 3T)	47/0, 1.4×10^{-14} (3W 0T)
CIC-IDS2017	XGBoost	40/0, 1.8×10^{-12} (5W 0T)	0/0, 1 (0W 5T)	264/0, 6.7×10^{-80} (5W 0T)
	LightGBM	535/0, 1.8×10^{-161} (3W 0T)	163/0, 1.7×10^{-49} (1W 2T)	410/0, 7.6×10^{-124} (3W 0T)
	CatBoost	14/0, 1.2×10^{-4} (3W 0T)	1/0, 1 (1W 2T)	122/0, 3.8×10^{-37} (3W 0T)
UNSW-NB15	XGBoost	602/10, 2.3×10^{-163} (5W 0T)	188/12, 8.1×10^{-42} (5W 0T)	998/4, 2.0×10^{-291} (5W 0T)
	LightGBM	373/1, 1.9×10^{-110} (3W 0T)	76/5, 2.3×10^{-17} (3W 0T)	612/1, 3.6×10^{-182} (3W 0T)
	CatBoost	243/3, 4.4×10^{-68} (3W 0T)	221/1, 6.6×10^{-65} (3W 0T)	611/1, 7.2×10^{-182} (3W 0T)

(FFE-v1 with $M = 5$ and $M = 20$) and (FFE-v1 with $M = 20$ and the white-box attacker); LGAT uses 5,000 FFE-v1 flows. The monotone-safe ratio features (backward/forward ratios) gave no gain in exploration and are not used. A manipulation-signature detector is implemented (`src/defenses/anomaly.py`) but was not evaluated, and random forests are supported by the recipe (the certificate test passes for them) but were not run. In exploration on validation, raising XGBoost's capacity from 60×6 to 300×8 lifted the monotone model's accuracy from 84.9% to 86.4% on UNSW-NB15 and from 99.85% to 99.94% on CIC-IDS2017; the follow-up below separates capacity from the constraint with an unconstrained 300×8 arm.

Selection. Table S11 gives the components chosen on validation. On four of the nine cells (CSE-CIC-IDS2018 LightGBM, CIC-IDS2017 CatBoost and LightGBM, UNSW-NB15 CatBoost) the selected free component of V2 is the undefended model itself, so the gain there comes entirely from the gate.

Cost of the gate. Table S7 gives the stand-alone gate's clean metrics relative to the undefended model of the same library, and the number of cohort flows (out of 1,000 detected by the undefended model) that the gate and V2 still detect.

Table S7. Stand-alone certified gate minus undefended model (accuracy, F1, recall; percentage points) and cohort flows still detected.

Corpus	**Library**	**Gate minus undefended**			**Gate kept**	**V2 kept**
		Acc.	F1	Rec.		
CSE-CIC-IDS2018	XGBoost	-0.70	-3.8	-5.4	917	998
	LightGBM	-0.70	-3.8	-5.8	921	1000
	CatBoost	-0.68	-3.7	-5.8	910	998
CIC-IDS2017	XGBoost	-0.03	-0.0	+0.0	998	999
	LightGBM	-0.05	-0.1	-0.1	999	1000
	CatBoost	+0.07	+0.1	+0.4	999	1000
UNSW-NB15	XGBoost	-1.77	-1.8	-1.3	955	998
	LightGBM	-1.56	-1.6	-0.8	951	992
	CatBoost	-0.08	-0.4	-2.2	940	1000

Tables S8–S10 give, for every corpus and library, clean accuracy, F1 and recall (%) and Real ASR (%) of FFE-v1 with $M = 5$ and $M = 20$, CAPGD and the white-box attacker for all nine models (undefended, MPS, LGAT, MPS+LGAT, orbit, adversarial training, the certified gate, V2 and V2-R). "Kept" is the number of cohort flows the model still detects; "Ret." is the share of the whole cohort still detected after the attack (minimum over attacks). Wilson intervals, validation tables and exact McNemar tests of V2 against every other model are in `results/defense_v2_report.md`. Wall-clock build times (shared, heavily loaded machine) put V2 at 5–33 times the undefended training time.

Follow-up checks (`results/defense_v2_followup.md`; models were retrained with the same seeds and reproduce the saved cohorts and clean accuracy exactly; Tables S12 and S13). *Whole-cohort evasion* is the share of the 1,000-flow cohort that is not flagged after the attack, counting flows the model already misses as evaded; for the undefended models it equals Real ASR. (i) *Hiding the backward columns.* The gate G35 additionally hides x_3 and x_5, which

Table S8. CSE-CIC-IDS2018: all defense models, all attacks.

Library	**Model**	**Acc.**	**F1**	**Rec.**	**FFE5**	**FFE20**	**CAPGD**	**WB**	**Kept**	**Ret.**
XGBoost	Undefended	97.30	87.3	77.6	73.3	99.5	99.0	99.5	1000	0.5
	MPS	97.30	87.3	77.6	61.9	61.9	98.3	99.5	999	0.5
	LGAT	97.28	87.3	77.8	25.6	73.1	96.1	99.5	998	0.5
	MPS+LGAT	97.30	87.3	77.6	45.5	45.5	95.3	99.5	998	0.5
	Orbit	97.28	87.2	77.8	9.5	20.4	3.9	81.5	998	18.5
	AdvTrain	97.27	87.2	77.8	1.4	9.0	6.8	80.7	998	19.3
	Gate (certified)	96.60	83.6	72.2	0.0	0.0	0.0	0.0	917	91.7
	V2	97.29	87.4	78.4	8.1	8.1	0.3	8.1	998	91.7
	V2-R	97.29	87.4	78.4	8.1	8.1	0.3	8.1	998	91.7
LightGBM	Undefended	97.28	87.3	77.8	68.4	99.8	99.5	99.8	1000	0.2
	MPS	97.28	87.3	77.7	42.3	42.3	98.5	99.8	998	0.2
	LGAT	97.28	87.2	77.8	12.8	56.8	96.3	99.8	1000	0.2
	MPS+LGAT	97.29	87.3	77.7	38.3	38.3	98.7	99.7	998	0.3
	Orbit	97.27	87.3	78.1	8.8	12.2	9.4	81.6	1000	18.4
	AdvTrain	97.27	87.2	78.0	1.3	8.7	7.4	81.5	1000	18.5
	Gate (certified)	96.58	83.4	72.0	0.0	0.0	0.0	0.0	921	92.1
	V2	97.27	87.3	78.2	7.9	7.9	0.2	7.9	1000	92.1
	V2-R	97.27	87.3	78.2	7.9	7.9	0.2	7.9	1000	92.1
CatBoost	Undefended	97.26	87.2	77.6	56.1	59.8	99.3	100.0	1000	0.0
	MPS	97.28	87.2	77.6	39.8	39.8	99.2	100.0	998	0.0
	LGAT	97.25	87.1	77.5	5.5	45.6	83.8	100.0	998	0.0
	MPS+LGAT	97.22	87.0	77.6	3.2	3.2	99.3	100.0	999	0.0
	Orbit	97.24	87.0	77.4	1.6	3.1	49.9	80.2	998	19.8
	AdvTrain	97.25	87.1	77.4	1.8	3.1	46.6	80.0	998	20.0
	Gate (certified)	96.58	83.4	71.9	0.0	0.0	0.0	0.0	910	91.0
	V2	97.25	87.2	77.9	0.0	0.0	0.0	8.8	998	91.0
	V2-R	97.25	87.2	77.9	0.0	0.0	0.0	8.8	998	91.0

Table S9. CIC-IDS2017: all defense models, all attacks.

Library	**Model**	**Acc.**	**F1**	**Rec.**	**FFE5**	**FFE20**	**CAPGD**	**WB**	**Kept**	**Ret.**
XGBoost	Undefended	99.96	99.9	99.8	69.7	99.1	77.0	100.0	1000	0.0
	MPS	99.91	99.8	99.7	92.8	92.8	47.7	98.1	999	1.9
	LGAT	99.96	99.9	99.9	3.0	31.9	52.6	100.0	1000	0.0
	MPS+LGAT	99.92	99.8	99.7	19.7	32.5	57.4	94.9	999	5.1
	Orbit	99.95	99.9	99.8	2.1	2.8	7.4	34.3	1000	65.7
	AdvTrain	99.95	99.9	99.8	1.6	2.6	1.2	15.2	1000	84.8
	Gate (certified)	99.93	99.9	99.8	0.0	0.0	0.0	0.0	998	99.8
	V2	99.97	99.9	99.9	0.2	0.7	0.7	0.7	999	99.2
	V2-R	99.97	99.9	99.9	0.2	0.7	0.7	0.7	999	99.2
LightGBM	Undefended	99.96	99.9	99.9	20.3	34.6	57.9	35.7	1000	42.1
	MPS	99.91	99.8	99.7	34.1	34.1	31.6	35.4	999	64.5
	LGAT	99.96	99.9	99.9	22.7	33.4	36.3	35.6	1000	63.7
	MPS+LGAT	99.92	99.8	99.7	33.3	33.4	95.9	35.3	999	4.1
	Orbit	99.94	99.9	99.8	2.0	4.0	3.1	35.4	1000	64.6
	AdvTrain	99.94	99.9	99.8	1.8	2.8	2.3	18.5	1000	81.5
	Gate (certified)	99.91	99.8	99.8	0.0	0.0	0.0	0.0	999	99.9
	V2	99.97	99.9	99.9	0.4	0.4	0.4	0.4	1000	99.6
	V2-R	99.97	99.9	99.9	0.4	0.4	0.4	0.4	1000	99.6
CatBoost	Undefended	99.85	99.7	99.4	31.1	32.0	5.4	33.6	1000	66.4
	MPS	99.82	99.6	99.4	32.0	32.0	25.0	32.8	997	67.0
	LGAT	99.86	99.7	99.5	2.1	31.2	31.0	33.2	1000	66.8
	MPS+LGAT	99.83	99.6	99.4	20.2	31.7	26.8	33.2	996	66.5
	Orbit	99.80	99.6	99.3	1.4	1.4	21.6	32.3	999	67.6
	AdvTrain	99.84	99.7	99.5	1.3	1.6	10.1	32.3	998	67.6
	Gate (certified)	99.92	99.8	99.8	0.0	0.0	0.0	0.0	999	99.9
	V2	99.93	99.8	99.9	0.1	0.1	0.1	0.1	1000	99.9
	V2-R	99.93	99.8	99.9	0.1	0.1	0.1	0.1	1000	99.9

Table S10. UNSW-NB15: all defense models, all attacks.

Library	Model	Acc.	F1	Rec.	FFE5	FFE20	CAPGD	WB	Kept	Ret.
XGBoost	Undefended	88.51	87.3	88.0	49.0	62.2	24.4	90.9	1000	9.1
	MPS	87.80	86.4	86.4	33.0	33.6	16.3	45.3	963	52.7
	LGAT	87.75	87.2	92.3	9.8	15.0	6.2	81.7	996	18.2
	MPS+LGAT	87.44	86.6	90.1	7.4	7.4	7.3	32.4	973	65.8
	Orbit	87.79	87.0	91.1	8.6	12.3	3.6	18.4	992	80.9
	AdvTrain	86.60	86.3	93.7	5.2	6.5	3.7	14.6	991	84.6
	Gate (certified)	86.74	85.5	86.7	0.0	0.0	0.0	0.0	955	95.5
	V2	88.38	87.5	90.4	9.6	12.2	1.6	25.4	998	74.5
	V2-R	86.38	86.2	94.8	2.1	2.1	0.6	2.6	994	96.8
LightGBM	Undefended	88.47	87.3	87.9	52.9	66.4	32.7	89.5	1000	10.5
	MPS	87.67	86.2	85.8	29.9	31.2	10.7	48.9	952	48.6
	LGAT	87.77	87.1	91.7	9.7	14.5	6.3	72.3	995	27.6
	MPS+LGAT	87.27	86.5	90.9	9.0	9.4	4.0	36.5	983	62.4
	Orbit	87.48	86.6	90.1	10.1	12.3	6.2	34.4	985	64.6
	AdvTrain	86.09	85.8	93.7	4.3	4.6	3.3	6.0	989	93.0
	Gate (certified)	86.91	85.7	87.2	0.0	0.0	0.0	0.0	951	95.1
	V2	88.09	87.2	90.5	19.6	22.0	2.1	31.9	992	67.6
	V2-R	86.31	86.1	94.3	1.3	1.3	0.6	1.3	994	98.1
CatBoost	Undefended	86.67	85.8	89.6	62.2	72.6	27.5	60.1	1000	27.4
	MPS	84.71	84.0	89.3	61.6	61.6	27.5	65.2	974	33.9
	LGAT	86.20	85.9	93.6	17.2	27.1	9.0	45.4	993	54.2
	MPS+LGAT	85.18	85.0	93.1	20.8	22.2	7.1	45.8	987	53.5
	Orbit	84.08	84.1	93.8	14.7	19.0	10.9	40.7	982	58.2
	AdvTrain	84.06	84.3	95.0	8.8	9.1	12.4	37.5	990	61.9
	Gate (certified)	86.59	85.4	87.4	0.0	0.0	0.0	0.0	940	94.0
	V2	86.42	86.0	92.6	4.9	4.9	1.0	5.7	1000	94.3
	V2-R	84.31	84.8	96.9	1.0	1.0	0.5	4.7	998	95.1

Table S11. Components selected on the validation split (free classifier, gate threshold τ) and their validation Real ASR (%).

Corpus	Library	V2 free	τ	val. FFE-v1	val. WB	V2-R free	τ
CSE-CIC-IDS2018	XGBoost	orbit light	0.5	6.0	6.0	orbit light	0.5
	LightGBM	undefended	0.5	8.0	8.0	undefended	0.5
	CatBoost	orbit light	0.5	0.0	8.4	orbit light	0.5
CIC-IDS2017	XGBoost	orbit light	0.95	0.3	0.7	orbit light	0.95
	LightGBM	undefended	0.8	0.0	0.0	undefended	0.8
	CatBoost	undefended	0.5	0.0	0.0	undefended	0.5
UNSW-NB15	XGBoost	orbit light	0.8	14.0	27.7	adversarial	0.5
	LightGBM	orbit light	0.8	21.4	29.8	adversarial	0.8
	CatBoost	undefended	0.5	4.0	4.3	adversarial	0.5

would remove the assumption that backward traffic does not change. It is costly: accuracy and F1 fall by about 4 and 19 points on CSE-CIC-IDS2018 and by 4.7–5.2 and about 9 points on CIC-IDS2017 relative to the undefended model, and by 12–14 accuracy points on UNSW-NB15; its benign false-positive rate is 2.1%, 5.9–6.1% and 36–37%, and it flags only 74–76% of the CSE-CIC-IDS2018 cohort. V2 rebuilt on G35 is much weaker where the gate matters: on CSE-CIC-IDS2018 its whole-cohort evasion is 35–36% against FFE-v1 ($M = 20$), 23–70% against CAPGD and 78–79% against the white-box attacker, against at most 9.0% for V2. (ii) *Capacity control.* An unconstrained 300×8 model with the gate's hidden columns is as accurate as the undefended model (CSE-CIC-IDS2018: 97.1% against 97.3%; UNSW-NB15: 89.2–89.4% against 86.7–88.5%) and equally evadable (whole-cohort evasion 17–98% across attackers on CSE-CIC-IDS2018). The gate's accuracy cost relative to this unconstrained arm (0.55 accuracy and 3.1 F1 points on CSE-CIC-IDS2018, 2.3–2.8 accuracy points on UNSW-NB15, 0.06 on CIC-IDS2017) is therefore the monotonicity itself, not a capacity effect. (iii) *False positives and drift.* Benign FPR on the test split is 0.11–0.14% for V2 on CSE-CIC-IDS2018 against 0.03–0.07% undefended, 0.02–0.05% against 0.01–0.02% on CIC-IDS2017, and 13–19% against 11–16% on UNSW-NB15; the FPR on benign flows in the top decile of forward packets is about the same as undefended, so we found no monotone drift for V2 or the gate. (iv) *FFE-v2 against the hardened models* was run only on CIC-IDS2017 XGBoost, first 300 cohort flows, retrained models, configuration of `results/ffev2_config.json` with snapping: Real ASR / whole-cohort evasion are 100.0% / 100.0% for the undefended model, 1.0% / 1.3% for V2, 8.4% / 8.7% for V2 with G35 and 0.0% / 2.3% for G35 alone. It was not run on other corpora or libraries. (v) *V2-R* was defined while building the pipeline, after a smoke run on UNSW-NB15 LightGBM (100-flow test cohort) in which its validation table and the undefended model's attack rates were seen, before any full test result existed; it is a post-hoc second tier and V2 (strict) is the headline. The selection rule is as in Section 7.1; the undefended model is V2's free component in four of the nine cells (CSE-CIC-IDS2018 LightGBM, CIC-IDS2017 LightGBM and CatBoost, UNSW-NB15 CatBoost).

Table S12. Clean accuracy (%), F1 (%) and benign false-positive rate (%) of the undefended model (U), the certified gate (G), the gate hiding x_3, x_5 (G35), the unconstrained 300×8 control (Unc.) and V2.

Corpus	Library	Acc. U	G	G35	Unc.	F1 U	G	G35	Unc.	FPR U	V2	G35
CSE-CIC-IDS2018	XGBoost	97.30	96.60	93.20	97.14	87.3	83.6	67.4	86.7	0.03	0.14	2.09
	LightGBM	97.28	96.58	93.39	97.15	87.3	83.4	68.5	86.7	0.07	0.14	2.09
	CatBoost	97.26	96.58	93.38	97.14	87.2	83.4	68.5	86.6	0.07	0.11	2.08
CIC-IDS2017	XGBoost	99.96	99.93	95.07	99.99	99.9	99.9	90.5	100.0	0.01	0.02	5.96
	LightGBM	99.96	99.91	94.76	99.99	99.9	99.8	89.9	100.0	0.01	0.03	6.12
	CatBoost	99.85	99.92	95.16	99.98	99.7	99.8	90.7	100.0	0.02	0.05	5.89
UNSW-NB15	XGBoost	88.51	86.74	74.38	89.30	87.3	85.5	75.4	88.0	11.07	13.30	36.08
	LightGBM	88.47	86.91	74.04	89.22	87.3	85.7	75.2	87.9	11.08	13.89	36.90
	CatBoost	86.67	86.59	74.36	89.39	85.8	85.4	75.3	88.1	15.70	18.62	36.00

Table S13. Whole-cohort evasion (%) of V2, V2 rebuilt on G35 and the unconstrained control against FFE-v1 ($M = 20$), CAPGD and the white-box attacker (WB).

Corpus	Library	V2			V2 with G35			Unc.		
		FFE	CAPGD	WB	FFE	CAPGD	WB	FFE	CAPGD	WB
CSE-CIC-IDS2018	XGBoost	8.3	0.5	8.3	36.3	50.4	79.0	86.1	17.4	95.2
	LightGBM	7.9	0.2	7.9	35.9	23.1	78.8	89.1	27.3	96.4
	CatBoost	0.2	0.2	9.0	35.1	69.6	77.7	93.2	30.5	98.1
CIC-IDS2017	XGBoost	0.8	0.8	0.8	1.3	1.6	5.5	32.4	5.0	33.9
	LightGBM	0.4	0.4	0.4	2.0	2.0	6.3	34.3	30.5	47.9
	CatBoost	0.1	0.1	0.1	1.1	2.8	3.1	32.2	0.7	32.5
UNSW-NB15	XGBoost	12.4	1.8	25.5	20.2	6.1	38.7	49.3	24.3	75.2
	LightGBM	22.6	2.9	32.4	58.5	15.2	81.6	58.4	31.0	86.8
	CatBoost	4.9	1.0	5.7	61.0	13.8	70.9	52.4	22.1	68.5

S7. White-box leaf-tuple-style control

The official LT-Attack code needs Bazel and a C++ compiler, which this machine lacks, so we wrote a greedy attack in the same spirit (`src/attacks/whitebox_tree.py`). For XGBoost it reads all 60 trees; each step moves one feature to the other side of one split threshold, choosing the move that lowers the exact ensemble margin most, for at most 20 steps (margins agree with `predict_proba` to 2×10^{-7}). It is not LT-Attack's leaf-tuple search with an L_∞ objective. Two variants share the projection and bookkeeping of the other attacks: *project at end* (unconstrained attack, then projection, then re-scoring) and *project in loop* (every candidate is projected before scoring). Both use all 1,000 cohort flows, two target queries (the tree reads are not counted; about 300–1,000 internal margin evaluations per flow) and are schema-feasible by construction. For LightGBM and CatBoost the control of Table 2 is our generic version on the raw ensemble margin (`RawMarginModel`), run with projection in the loop; its 56.8% on UNSW-NB15 CatBoost may reflect single-feature greedy steps stalling on oblivious trees, which we did not test.

Table S14. Real ASR (%) of the XGBoost white-box control and FFE-v1; exact McNemar p of the in-loop variant against FFE-v1 on the same 1,000 flows.

Corpus	Project at end	Project in loop	FFE-v1	p (in loop vs v1)
CSE-CIC-IDS2018	31.6	99.5	99.5	1
CIC-IDS2017	35.1	100.0	99.2	0.0078
UNSW-NB15	0.0	90.2	57.6	9.8×10^{-88}

Table S14 gives the XGBoost results. Before projection the attack evades every flow on all three corpora; most of that comes from the derived packet-rate columns and the immutable columns, which the projection undoes. The project-at-end variant is the fair version of an unconstrained white-box attack and is far below FFE-v1 ($p < 10^{-170}$ on UNSW-NB15). The in-loop variant equals FFE-v1 on CSE-CIC-IDS2018, beats it on the other two corpora, and is the reference against which FFE-v2 is judged in the main text. It sees the trees and is cost-blind: its median operational cost is 213 against 40.0 for FFE-v2 and 5.6 for FFE-v1

on UNSW-NB15 XGBoost.

S8. Packet-level replay

We built a synthetic replay (`src/validation/packet_replay.py`); no packet capture was used and nothing was sent over a network. A flow record is expanded into a packet sequence with the stated packet counts, payload and header totals, duration and initial window, and a CICFlowMeter-style extractor (our re-implementation, not CICFlowMeter) recomputes the twelve features. For 20,000 flows per corpus the round trip is exact for every flow on CSE-CIC-IDS2018 and CIC-IDS2017 and for 19,999 of 20,000 on UNSW-NB15 (one one-packet record with a duration of 1 μs); this shows that the schema is jointly realizable by packet lists, not that it matches a real capture.

For FFE-v2 the integer action ledger of each successful flow (delay, chaff, re-segmented packets, padding, window) is applied to the trace; segmentation halves the largest forward packet K times, and tests confirm that payload, duration, first forward packet and window are unchanged. The generators book 28–52 B of forward payload per chaff packet, whereas a minimal packet carries none, so three conventions are replayed: *schema* (chaff carries the booked bytes), *empty chaff, bytes as padding* (empty chaff, same feature vector), and *empty chaff, bytes dropped* (the booked bytes are not delivered). Table S15 gives the share of schema-level evasions that remain evasive (XGBoost, first 300 flows per corpus). Under the first two conventions 100% of replayed flows reproduce the schema vector within 1% on all twelve features and stay feasible, with no unmet segmentation or padding. The header-length mismatch of FFE-v1 does not arise because the ledger, not the vector difference, drives the replay.

Table S15. Share (%) of FFE-v2's schema-level evasions that remain evasive after packet-level replay (XGBoost).

Corpus	Flows	Evasions	Schema	Empty, as padding	Empty, dropped	Segmenting flows
CSE-CIC-IDS2018	300	298	100.0	100.0	100.0	0
CIC-IDS2017	300	300	100.0	100.0	100.0	0
UNSW-NB15	300	295	100.0	100.0	69.2	118

For FFE-v1 on the same flows, the earlier replay (`results/packet_replay.md`) found 99.3%, 100.0% and 100.0% retained under the first two conventions and 100.0%, 99.0% and 68.2% when the booked bytes are dropped, and a header-length mismatch on 9.1%, 2.3% and 6.1% of the successful flows. UNSW-NB15 depends on the booked chaff bytes because most of its successful flows use chaff and padding. The replay does not model a TCP state machine, server reaction (including additional ACKs elicited by more forward segments), offload or flow timeouts, and part of its agreement holds by construction because the action magnitudes are derived from the adversarial vector. It does not test whether an attack payload still works after dilation, padding or segmentation.

S9. Why LightGBM resisted FFE-v1 on CIC-IDS2017, and the fix

On the 1,000-flow cohort FFE-v1 evaded XGBoost in 99.2% and LightGBM in 35.1%. By family, LightGBM was evaded in 100% of port scans, botnet and brute-force flows but in 1.9% of DoS and 0.0% of DDoS flows, about 66% of the cohort; XGBoost was evaded in 98.1% and 100%. Threshold extraction and the probe are correct for LightGBM, start scores are identical (median 0.998) and the gain profiles are nearly the same (backward payload 0.31 for both, duration 0.29 for both), so reliance on immutable features does not explain the gap. Setting only the initial window to 64,240 B, FFE-v1's target, flips 97.2% of the 33,857 detected DoS test flows for XGBoost and 0.9% for LightGBM (99.96% against 0.0% for DDoS). Almost no training flow has a window between 29,200 and 65,535 B (64,240 B itself occurs in 0.08% of benign and 0.015% of attack TCP flows; `results/diagnosis_lightgbm.md`); XGBoost splits at 32,831 and 65,535, so 64,240 B lands on the benign side, whereas LightGBM splits at 63,586 and 65,435 and keeps it on the malicious side. More leaves, finer histogram bins and a smaller minimum leaf size change this by about one point. Moving only the window target to the in-support value 65,535 B lifted FFE-v1 on LightGBM from 35.1% to 93.7% (queries 120 to 21.8) and left XGBoost at 99.2%. FFE-v2 replaces the single target by the attacker's most frequent benign windows and reaches 96.0% on LightGBM with 32.4 queries; the residual failures are concentrated in DoS flows (90.5% evaded) and were not examined further.

S10. FFE-v1 ablations

These results refer to FFE-v1 (four generators, tree-split boundary snapping) and are kept as the history of the baseline. Table S16 compares FFE-v1 with and without snapping (TSBS) on the 1,000-flow cohorts: snapping leaves Real ASR unchanged on CSE-CIC-IDS2018, lifts it from 97.5% to 99.2% on CIC-IDS2017 and from 54.6% to 57.6% on UNSW-NB15, and costs up to 1.8 times as many queries. Table S17 reports step-budget, generator-subset and cost-regime ablations on the CUDA XGBoost target of CSE-CIC-IDS2018 with a balanced cohort of 283 detected malicious flows (up to 50 per family): success rises from 42.8% at $M = 1$ to 98.9% at $M \geq 10$; no single generator suffices (10.2%–66.1%); the cost regime and snapping offset leave 98.9% unchanged on this saturated corpus. Table S18 gives clean performance and surrogate agreement of FFE-v1's targets.

S11. Cross-architecture transferability of FFE-v1

Flows crafted by FFE-v1 against a source ensemble were scored by the three other architectures (Table S19). The cohort is the 221 CSE-CIC-IDS2018 malicious flows (up to 50 per family) that XGBoost, LightGBM, CatBoost and Random Forest all detect when clean, all trained on the same partition. Flows crafted against XGBoost evade LightGBM in 71.5%, Random Forest in 55.7% and CatBoost in 24.0%; LightGBM flows evade XGBoost in 75.6%, Random Forest in 63.8% and CatBoost in 24.0%; CatBoost flows evade XGBoost in 70.1%,

Table S16. FFE-v1 with and without tree-split boundary snapping (XGBoost, Real ASR with Wilson 95% interval).

Dataset	**Variant**	**Real ASR (%)**	**Queries**	**ms/flow**
CSE-CIC-IDS2018	FFE	99.5 [98.8, 99.8]	16.5	15.8
	FFE without TSBS	99.5 [98.8, 99.8]	11.6	10.6
CIC-IDS2017	FFE	99.2 [98.4, 99.6]	14.6	10.7
	FFE without TSBS	97.5 [96.3, 98.3]	13.5	9.6
UNSW-NB15	FFE	57.6 [54.5, 60.6]	93.7	89.1
	FFE without TSBS	54.6 [51.5, 57.7]	51.5	36.0

Table S17. Ablation of FFE-v1 on CUDA XGBoost, CSE-CIC-IDS2018 ($N = 283$). Delay is the mean added flow duration.

Configuration	**ASR (%)**	**PFR (%)**	**Queries**	**Added delay (s)**
Panel A: step budget M				
$M = 1$	42.8	100.0	5.0	0.08
$M = 2$	63.3	100.0	8.3	0.14
$M = 3$	67.1	100.0	9.5	0.15
$M = 5$	84.1	100.0	12.2	0.16
$M = 10$	98.9	100.0	13.9	0.18
$M = 20$	98.9	100.0	14.4	0.18
$M = 50$	98.9	100.0	16.0	0.18
Panel B: generator subset ($M = 20$)				
Temporal Dilation Only	66.1	100.0	32.6	40.72
Payload Padding Only	10.2	100.0	5.1	0.00
Chaff Injection Only	32.5	100.0	9.7	0.00
Window Morphing Only	46.3	100.0	2.5	0.00
Time + Padding	69.3	100.0	39.6	26.81
Time + Chaff	78.1	100.0	38.4	22.05
Padding + Chaff	33.6	100.0	14.2	0.00
Complete Lie Cone	98.9	100.0	14.4	0.18
Panel C: operational cost regime ($M = 20$)				
A (Default / Balanced)	98.9	100.0	14.4	0.18
B (Latency-Sensitive)	98.9	100.0	14.1	0.13
C (Unconstrained Delay)	98.9	100.0	14.4	0.18
D (Uniform / Flat)	98.9	100.0	14.3	0.14

Table S18. FFE-v1 against XGBoost and LightGBM: clean test performance, Real ASR on the 1,000-flow cohort (Wilson 95% intervals) and surrogate–target agreement.

Dataset	Model	Acc. (%)	F1 (%)	AUC (%)	FFE Real ASR (%)	PFR (%)	Queries	Surr. agr. (%)
CSE-CIC-IDS2018	XGBoost	97.30	87.31	96.81	99.5 [98.8, 99.8]	100.0	16.5	99.8
	LightGBM	97.28	87.28	96.77	99.5 [98.8, 99.8]	100.0	20.2	99.7
CIC-IDS2017	XGBoost	99.96	99.91	100.00	99.2 [98.4, 99.6]	100.0	14.6	99.9
	LightGBM	99.96	99.92	100.00	35.1 [32.2, 38.1]	100.0	120.1	99.9
UNSW-NB15	XGBoost	88.51	87.32	96.47	57.6 [54.5, 60.6]	100.0	93.7	87.6
	LightGBM	88.47	87.28	96.43	62.4 [59.4, 65.3]	100.0	89.0	87.4

LightGBM in 77.4% and Random Forest in 77.4%; Random Forest flows evade XGBoost in 78.7%, LightGBM in 71.0% and CatBoost in 24.0%. CatBoost is the hardest transfer target, but direct score-only attacks on it succeed in 99.5%, 36.4% and 97.5% of the flows of the three corpora with FFE-v2 (Table 2), so the low transfer rate reflects mismatched borders rather than robustness.

Table S19. Transferability of FFE-v1 flows (success rate in %, $N = 221$; diagonal: attack against the source model itself).

Source \ Target	XGBoost (CUDA)	LightGBM	CatBoost	Random Forest
XGBoost (CUDA)	**100.0**	71.5	24.0	55.7
LightGBM	75.6	**100.0**	24.0	63.8
CatBoost	70.1	77.4	**69.7**	77.4
Random Forest	78.7	71.0	24.0	**100.0**

S12. Corpora and cleaning

Table S20 gives the record counts. Valid records pass the validity filter; Kept = Valid − Duplicates, where duplicates are records with identical features and label after projection onto the manifold. Records with identical features but different labels are kept (149,526, 22 and 1,040). Cleaning drops records with missing or infinite values, non-positive duration, no forward packet or a protocol other than TCP or UDP, and TCP flows without an observed window (the window of UDP flows, recorded as −1 by CICFlowMeter, is set to 0); in CIC-IDS2017 the 11,974 "Attempted" probes are removed and "Infiltration – Portscan" is counted as port scanning; in UNSW-NB15 zero-duration flows are dropped. The projection onto the manifold changes a feature by more than 5% in 3.1% of the valid CSE-CIC-IDS2018 records, 4.7% of CIC-IDS2017 and none of UNSW-NB15.

Table S20. Full-scale benchmark corpora.

Dataset	Raw rows	Valid	Duplicates	Kept	Attacks	Train	Test
CSE-CIC-IDS2018	8,284,254	8,153,708	3,185,319	4,968,389	594,809	3,477,872	993,679
CIC-IDS2017	2,099,971	2,085,851	426,982	1,658,869	396,403	1,161,208	331,774
UNSW-NB15	257,673	215,689	92,181	123,508	55,559	86,456	24,703

S13. Related-work taxonomy

Table S21. Representative evasion, verification and defense paradigms for tabular and network telemetry. Physical feasibility reports the empirical PFR of this benchmark where it was evaluated.

Method	Target model	Engine	Queries	Coupled invariants	Physical feasibility
Kantchelian et al.	GBDT, random forest	MILP (worst case $2^{\mathcal{O}(DT)}$)	0 (white-box)	Unsupported (linear only)	Unconstrained
Zhang et al. (LT-Attack)	Tree ensembles	Discrete leaf-tuple search	$O(N_{\text{iter}})$	Not modeled	Unconstrained
Chen et al. (ZOO)	Black-box classifiers	Zeroth-order coordinate descent	80–130 (mean)	Not modeled	0.0–83.2% of evasive flows (empirical)
Chen et al. (HSJ)	Black-box decision	Boundary binary walk	47 (mean)	Not modeled	0.0% (empirical)
Lin et al. (IDSGAN); Zolbayar et al.	Black-box NIDS	Learned generator on a surrogate	2.4–4.5 after training	Repaired after generation	100.0% (after repair, our re-implementation)
Ghamizi et al. (MOEVA)	Constrained tabular	Multi-objective genetic search	226–445 (mean)	Heuristic penalty	59.4–100.0% of evasive flows (empirical)
Simonetto et al. (CAA/CAPGD)	Tabular models	Constrained APGD (surrogate) with MOEVA	3.2–16.2 (CAPGD, mean)	Repair operator	100.0% (after repair, our re-implementation)
PSO baseline after Han et al. (ours)	Black-box NIDS	PSO on interval and padding features	183–251 (mean)	Invariant projection	100.0% (projected, our addition)
Incer et al.	Malware detectors	Monotone classification (defense)	–	Monotone features	Certificate for add-only attackers
FFE-v2 (ours)	**XGBoost, LightGBM, CatBoost**	**Multi-scale Lie-generator steering, scores only**	**28–38 on easy corpora, 109 on the ceiling cell**	**Exact closed-form generators plus projection**	**100.0% (schema-feasible)**

Note: T trees, D depth. Empirical query counts and feasible shares of the evasive candidates are ranges over the three full-scale benchmarks against XGBoost. LT-Attack was not run; Section S7 reports a greedy leaf-tuple-style control. CAPGD and the NIDSGAN-style generator are our re-implementations.